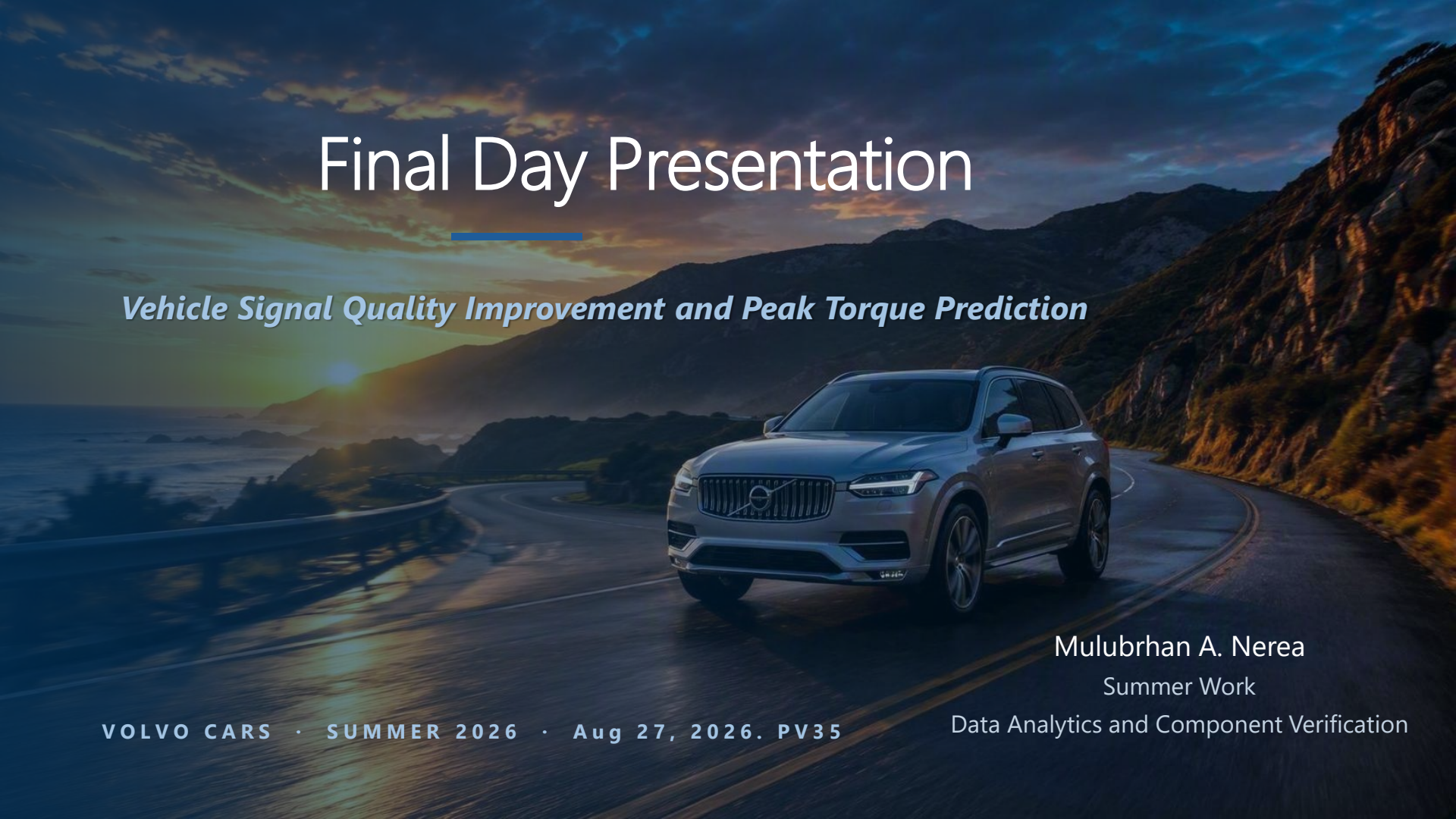
Final Day Presentation

Vehicle Signal Quality Improvement and Peak Torque Prediction



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Summer Work
Data Analytics and Component Verification

VOLVO CARS · SUMMER 2026 · Aug 27, 2026. PV35

What this presentation covers

01	**Context and problem**	Where the work sits and why measurement data needs cleaning
02	**Project 1 — Signal cleaning**	Rule-based and ML-based quality improvement for MDF/MF4 data
03	**Project 2 — Peak torque prediction**	Predicting future peak torque events from vehicle behaviour
04	**Results and key learnings**	What was delivered, and what I took away from it
05	**Summary and next steps**	Conclusions and what a future intern can pick up

Read this deck top to bottom — each section builds on the one before it.

Where this work sits

The team

Data Analytics & Component Verification at Volvo Cars. The department verifies that vehicle components behave as intended under real-world load; the analytics team turns measurement data from that testing into usable insight.

The data: MDF/MF4 measurement files

Instrumented vehicles are driven on test tracks and public roads. Driveline torque, wheel speed, temperature, voltage, current and other signals are logged at high frequency into MDF/MF4 files — large, multi-channel time-series datasets.

My two projects

1. **Make that measurement data trustworthy.**
2. Use vehicle behavior signals to predict peak torque events before they occur.



The problems I was asked to solve

Measurement data cannot be trusted as-is, and peak loads are only known after it has already been measured.

Signal-quality faults

MDF/MF4 files contain signal anomalies like offset faults, dropouts, invalid readings, and frozen-sensor behavior and abnormal measurements that corrupt any analysis built on top of them.

Cleaning was manual

Quality issues were handled case by case, per dataset and per engineer — slow, hard to reproduce, and waste time.

Many signals, many configurations

Torque & temperature faults occur differently. .
FWD, RWD and AWD vehicle configurations, two or Four signals to be cleaned.

Peak torque is only known afterwards

Peak load events are identified from measured torque for test vehicles, but in customer vehicles there is no way to anticipate them in advance.

PROJECT 1

VOLVO CARS

Vehicle signal cleaning and quality improvement

A batch-processing tool for automatic torque-signal cleaning and temperature-signal cleaning in vehicle measurement files, using both rule-based logic and machine learning.

Detect	Rule-based fault detection for spikes, stuck signals and out-of-range measurements, plus machine learning approaches for identifying abnormal signal behavior.
Repair	Signal repair through **interpolation, and substitution** from healthy reference channels where a faulty channel has a trustworthy equivalent.
Generalise	Started with wheel torque, then extended to temperature— including temperature-specific logic
Automate	Automated MDF/MF4 processing and output generation, with visualisation tools for comparing original and cleaned signals side by side.

Supports FWD, RWD and AWD vehicle configurations.

How the cleaning framework works

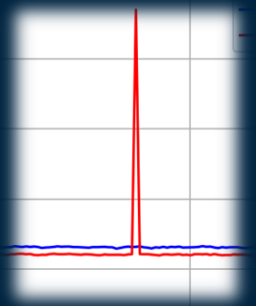
Setting	Value	Purpose
value_threshold	100 Nm	Starts paired-signal difference checking
window_size	20 samples	Detection window
activity_ratio	3.0	Requires one side to be much more dynamic
spike_jump	200 Nm	Sharp jump threshold
high_torque_threshold	1500 Nm	High-level plateau candidate
direct_step_abs_min	3000 Nm	Clear saturated step condition
step_mean_diff_threshold	500 Nm	Strong left-right difference
min_step_samples	5 samples	Minimum accepted step size
min_stuck_duration_s	20 s	Minimum duration for long stuck checking

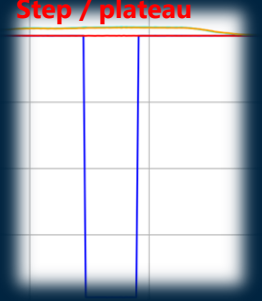
- **Load MDF/MF4**
  Batch-read .mdf, .mf4 and .dat measurement files and their channels automatically, no manual preparation per file.
- **Rule-based detection**
  Marks signal problems — bad values, frozen signals and broken sensors — using defined limits.
- **ML-based detection**
  Investigated machine learning methods to catch abnormal signal behavior that rule based method misses.
- **Repair**
  Interpolate short faults; substitute from a healthy reference channel where one is available.
- **Visualise and export**
  Plot original against cleaned signals for review, then export the cleaned dataset.

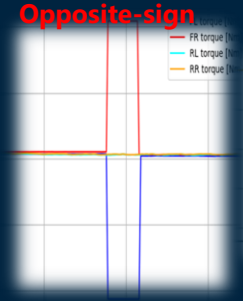
Built in Python as reusable modules, so the same framework runs on any new measurement campaign.

The torque faults the framework detects











- **Flat-top / plateau**
  Torque holds an unrealistically constant value where the signal should still be varying — a stuck or saturated channel.
- **Point spike**
  Single-sample jumps far outside the surrounding trend, caught through derivative and slope analysis.
- **Step / plateau event**
  Box-like level shifts where torque jumps to a new value, holds it, then returns.
- **Opposite-sign fault**
  Left and right wheel channels report opposite signs where the expert says they should agree.
- **Short plateau / micro-step**
  Brief, thin plateaus that are easy to miss by eye but still distort downstream analysis.

A hybrid rule-based signal-processing approach: derivative and slope analysis, threshold-based fault detection, and interpolation-based correction.

How faults are repaired, and what comes out

- **Self-interpolation**

  Short faults are reconstructed from the healthy samples on either side of the event.

- **Opposite-channel substitution**

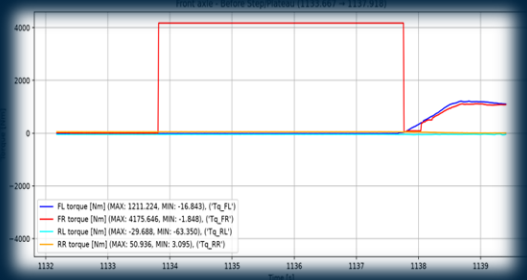
  Longer or unrecoverable faults are replaced using the corresponding healthy left or right axle channel.

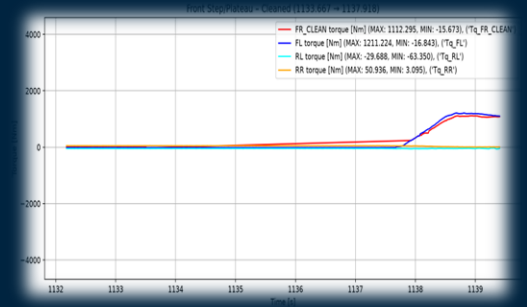
- **Fault-side decision**

  Logic decides which of the two channels is the faulty one before anything is overwritten.

- **Outputs for review**

  Per-fault before/after zoom plots, full-range comparison plots, a zoomed MF4 export and an Excel log of every cleaned event.





Runs as a batch process over .mdf, .mf4 and .dat files, on front and rear axle torque channels, for AWD, FWD and RWD vehicles.

One tool, two cleaning modes

- **AUTO mode — torque cleaning**
  Cleans TQ_FL, TQ_FR, TQ_RL and TQ_RR across FWD, RWD and AWD vehicles, detecting spikes, steps, plateaus, short pulses and frozen torque signals.
- **CUSTOM mode — temperature cleaning**
  Cleans manually selected air and bushing rubber temperature signals (FL, FR, RL, RR), catching impossible values such as -60 °C as well as smaller downward dropouts.
- **Repair logic per mode**
  Torque uses interpolation or substitution depending on the fault type. Temperature uses same-signal interpolation, with paired-signal substitution only when a signal is mostly invalid, set by an invalid-ratio threshold.
- **Output behaviour**
  Saves filename_Edited.mf4 when cleaning occurs, or the original filename.mf4 when no fault is found, plus original and cleaned full-range plots and an Excel fault log.

A batch-processing tool for automatic torque-signal cleaning and temperature-signal cleaning in MDF, MF4 and DAT vehicle measurement files.

How the tools work

One root-folder workflow with AUTO and CUSTOM processing paths

1 Run the .py file → **2 Select root folder** → **3 Detect and clean** → **4 Save and review**

AUTO mode — torque workflow

- Path text identifies FWD, RWD or AWD
- Front: TQ_FL compared with TQ_FR
- Rear: TQ_RL compared with TQ_RR
- Common paired motion is preserved
- Interpolation first, substitution when required

CUSTOM mode — temperature workflow

- Choose FWD, RWD or AWD manually
- Select two or four paired channels
- Valid range: 15 °C to 120 °C
- Local dip: 1.2 °C, 10 s window, 3 s padding
- Mostly invalid signal: substitute from valid pair

Output folder: <Selected_Root>_Edited | the original measurement folder is preserved

The user selects one root folder — the tool walks every measurement file inside it.

PROJECT 2

VOLVO CARS

Peak torque event prediction

> Predict upcoming peak torque events from vehicle behaviour signals — before they occur — to flag high driveline loading conditions in advance.

The idea

Peak torque events are moments of unusually high driveline loading. They are visible in measured torque. If a model can learn the vehicle behavior that leads up to them, those events can be anticipated instead of discovered.

Why it matters

The model predicts where high-torque events are likely to occur, helping engineers analyze large vehicle measurement datasets more efficiently and validate driveline loading conditions.



The machine learning pipeline

01
Dataset

Automated dataset generation from large volumes of MDF/MF4 vehicle measurement files.

02
Labels

Peak torque events identified from measured torque signals, with event windows created around them.

03
Features

Feature engineering from vehicle behaviour signals only — describing how the vehicle is being driven.

04
Model

XGBoost models trained and evaluated to predict whether a peak torque event is coming.

05
Validate

Deployment testing on unseen vehicle data and feature importance evaluation.

Design principle: no measured torque as an input feature

The model had to learn from vehicle behavior, not from the answer. Measured torque was used only to generate labels and to validate predictions — never as a model input.

PROJECT 2 · MACHINE LEARNING

VOLVO CARS

Peak Torque Predictor

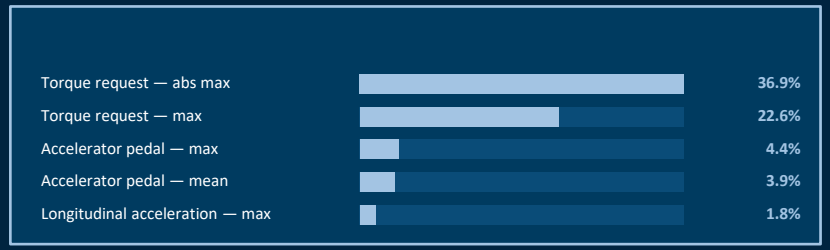
Behavior-only ML pipeline · strongest features · AWD validation

HOW IT WORKS

1	2	3	4	5
Peak labels	**5 s windows**	**Features**	**XGBoost**	**Events**
> 1800 Nm	2.5 s step	behavior only	probability	final validation

Model trained on vehicle behavior signals only — no measured torque.

TOP MODEL DRIVERS



Torque-request features contribute 61.9% of total importance.

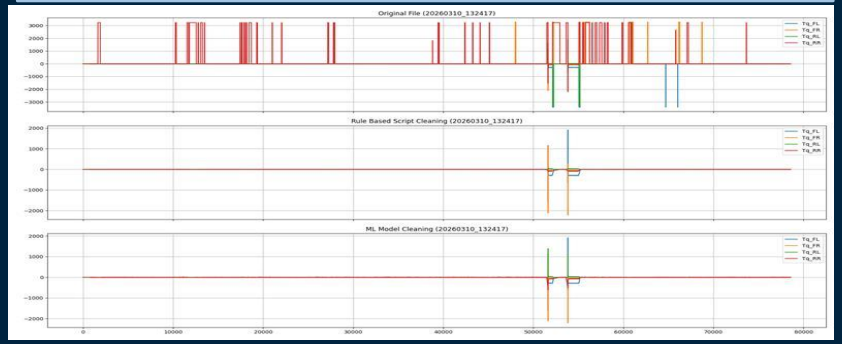
KEY PRINCIPLE **Measured wheel torque is used for labels and validation, never as an ML input.** ML: > 1 peak in 5 s | Event: ≥ 2 peaks > 1800 Nm within 5 s

AWD VALIDATION RESULTS

REAL EVENTS	PREDICTED	TP	FP	FN	SENSITIVITY	PRECISION	F-SCORE	Correlation Factor
162	241	129	112	33	79.6%	53.5%	64.0%	0.67

The model detected 129 of 162 real events using vehicle behavior only. | All selected AWD driving profiles.

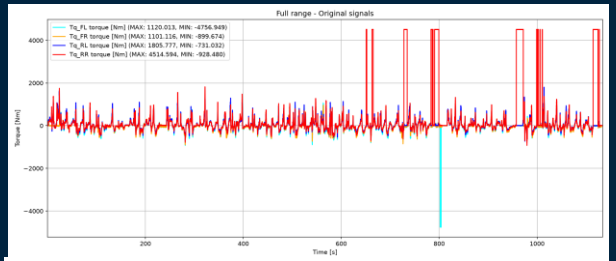
What was delivered

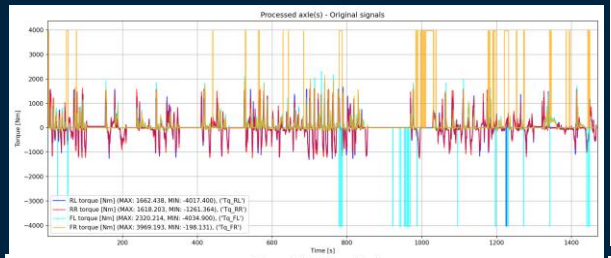


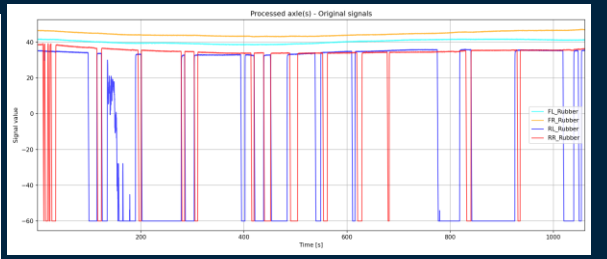
Project 1

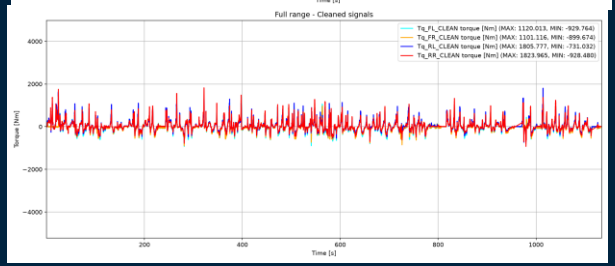
- Reusable signal-quality framework for MDF/MF4 data
- Rule-based and ML-based fault detection
- Coverage for torque, temperature, Repair via interpolation and healthy reference channels

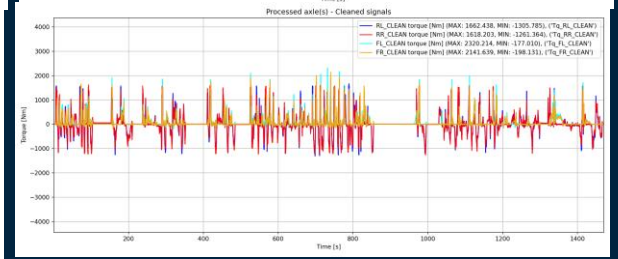
Temperature-Rubber/Air

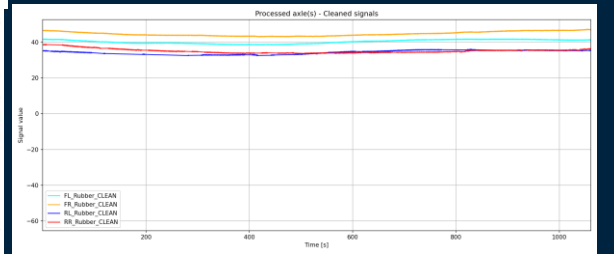






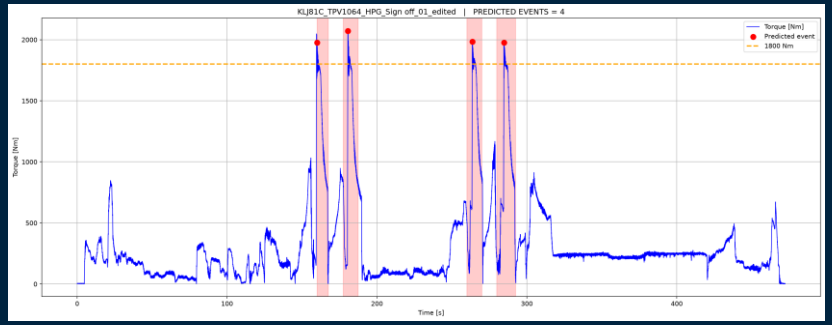


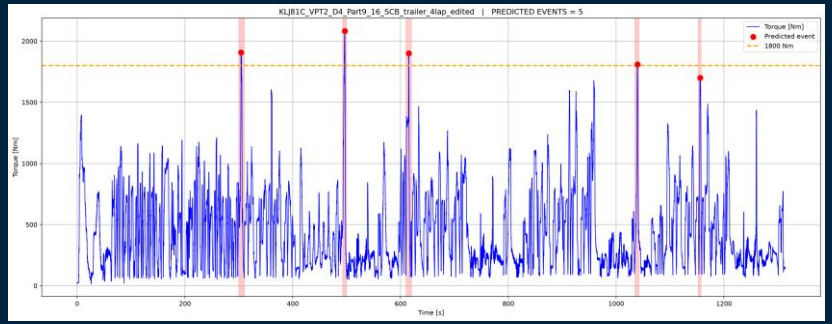




What was delivered

Peak torque prediction on unseen vehicle data

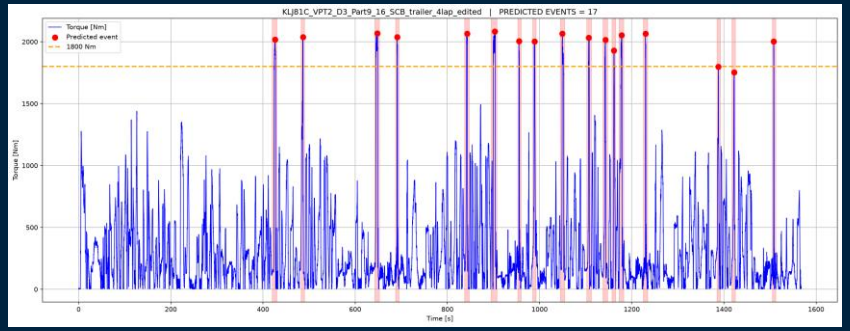




Project 2

PIPELINE

- End-to-end peak torque prediction pipeline, one vehicle sample
- Behavior-based features with no torque leakage
- XGBoost models validated on unseen vehicle data



What I learned

Technical

- Most of the value came from data quality and feature engineering, Learned had a larger impact than model selection.
- Preventing torque errors into the features mattered more than tuning XGBoost.

Domain

- Learned how driveline loading and peak torque events are interpreted, and how measurement campaigns actually work.
- Learned to validate ML predictions against real measurement data

Automotive industry

- Results must be traceable and physically explainable, not just statistically strong.
- Understood driveline loading and Signal behavior in AWD/FWD/RWD vehicles

Working at Volvo Cars

- An open culture where asking domain experts.
- Regular reviews kept the framework aligned with how Experts work with the data.
- Learned how engineering decisions are supported by measurement data.

Personal

- Learned to scope ambiguous problems myself and to be clear about what the data can and cannot support — stronger independent problem solving and ownership.
- Became more confident handling large and complex datasets.

Conclusion

Better measurement data quality and behaviour-based prediction together support more reliable vehicle analysis, engineering validation, and predictive analytics at Volvo Cars.

Conclusions

- Signal-quality faults can be detected and repaired systematically, not case by case
- Peak torque events can be predicted from vehicle behavior alone

Open follow-ups

- Extend ML-based fault detection beyond the rule-based baseline
- Add more labelled events to strengthen the prediction model
- Validate across more vehicle variants and driving conditions

Starting point for a future intern: the cleaning framework modules, the prediction pipeline, and the handover notes.

THANK YOU

Thank you, Teams,

To the Data Analytics team and Component Verification
Special thanks to Nitin, Jane and Eva.

Mulubrhan A. Nerea